AI Document Processing System

About the Project

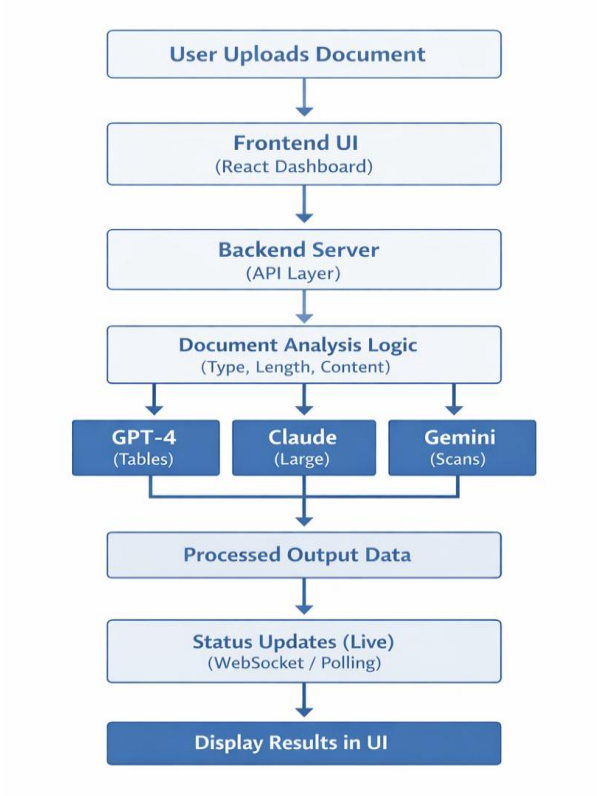
This project is a full-stack web application built to make document processing easier using AI. Instead of manually going through large documents, users can simply upload files and let the system analyze and

useful information.

The system also provides real-time updates so users can track what's happening with their document — whether it's still proc

essing or completed.

Pipeline Diagram



How It Works

1. User uploads a document through the frontend

2. The request is sent to the backend server
3. Backend analyzes the document (type, length, content)
4. Based on conditions, the system selects an AI model:
    - GPT-4 → for table extraction
    - Claude → for large documents
    - Gemini → for scanned PDFs
5. The document is processed and output is generated
6. Status updates are sent in real-time (WebSocket/Polling)
7. Final results are displayed in the UI

How to Run the Project

Step 1: Download the Project

Download this repository.

Step 2: Setup Backend

- Open the `backend` folder

▶ *Install Dependencies*

```
create a virtual environment

pip install -r requirements.txt
```

▶ *Run Backend Server*

In terminal type : uvicorn main:app --reload

Step 3: Setup Frontend

- Open the `frontend` folder

▶ *Install Dependencies*

```
npm install
```

▶ *Start Frontend*

```
npm start
```

Step 4: Open in Browser

Make sure backend is running, then open: